

# Imad EL BADISY | Resume

Research Engineer in Biostatistics

Position: Research Engineer in Biostatistics, CM6RI — Rabat, Morocco  
Field: Survival Analysis • Missing Data • Interpretable ML • Scientific Computing  
Education: Ph.D. Candidate, Aix-Marseille University (2021–2025)

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## Profile

Research Engineer in Biostatistics working at the interface of computational biostatistics and machine learning. My work focuses on developing reproducible analytical workflows, supporting biomedical research projects, and building methodological tools for health data science. I develop open-source R packages for survival analysis, missing data, and interpretable machine learning, and contribute as a consultant and trainer in biostatistics and applied data science.

## Experience

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
| Research Engineer in Biostatistics - Mohammed VI Center for Research and Innovation (CM6RI) Rabat, Morocco                                                                                                                                                                                                                                                                                                                                                           | Nov. 2021 | Present   |
| <ul style="list-style-type: none"><li>* Contributing to the structuring of an emerging AI and Data Science service</li><li>* Conducting methodological and applied research in health (oncology, fertility, biomedical domains)</li><li>* Providing methodological support in computational biostatistics, statistical modeling, and machine learning</li><li>* Developing reproducible workflows for health data science and biomedical research projects</li></ul> |           |           |
| Data Science Education Officer - The GRAPH Network Switzerland / Remote                                                                                                                                                                                                                                                                                                                                                                                              | Sep. 2023 | Dec. 2023 |
| <ul style="list-style-type: none"><li>* Designed and delivered training programs in data science for global health</li><li>* Developed educational material and course content for data science learners</li><li>* Contributed to capacity building in applied data science within a public health context</li></ul>                                                                                                                                                 |           |           |
| Research Engineer in Applied Statistics - Institut Mines-Télécom, IMT Atlantique Brest, France                                                                                                                                                                                                                                                                                                                                                                       | Sep. 2019 | Sep. 2021 |
| <ul style="list-style-type: none"><li>* Conducted statistical analyses of national health data (SNDS-based analyses)</li><li>* Implemented automated and reproducible data workflows</li><li>* Supported applied research in health data, epidemiology, and statistical modeling</li></ul>                                                                                                                                                                           |           |           |
| Biostatistician - INSERM Clermont-Ferrand, France                                                                                                                                                                                                                                                                                                                                                                                                                    | Jan. 2019 | Jun. 2019 |
| <ul style="list-style-type: none"><li>* Conducted cost-effectiveness evaluation of a clinical protocol on ketamine for chronic pain</li><li>* Contributed to statistical and health-economic analyses in a clinical research setting</li></ul>                                                                                                                                                                                                                       |           |           |
| Health Economist Intern - Université de Sherbrooke Sherbrooke, Quebec, Canada                                                                                                                                                                                                                                                                                                                                                                                        | Feb. 2017 | Sep. 2017 |
| <ul style="list-style-type: none"><li>* Conducted a systematic review and meta-analysis on health utilities in cancer patients</li><li>* Contributed to evidence synthesis and health economics research</li></ul>                                                                                                                                                                                                                                                   |           |           |

## Education

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|
| Ph.D. Candidate in Biostatistics - Aix-Marseille University, SESSTIM Marseille, France                                                                      | 2021 | 2025 |
| <ul style="list-style-type: none"><li>* Thesis: Missing Data and Machine Learning Methods for Survival Analysis</li><li>* Supervisor: Roch Giorgi</li></ul> |      |      |
| Master 2 in Quantitative Methods for Health Research - Aix-Marseille University, SESSTIM Marseille, France                                                  | 2018 | 2019 |
| <ul style="list-style-type: none"><li>* Advanced training in biostatistics, epidemiology, and quantitative methods for health research</li></ul>            |      |      |
| Master 1 in Applied Mathematics and Statistics - Université Clermont Auvergne Clermont-Ferrand, France                                                      | 2017 | 2018 |
| <ul style="list-style-type: none"><li>* Training in applied statistics, mathematical modeling, and data analysis</li></ul>                                  |      |      |
| Master in Health Economics - Université Clermont Auvergne, CERDI Clermont-Ferrand, France                                                                   | 2017 |      |

\* Specialization in health econometrics and economic evaluation

## Software (R Packages)

|                                                                      |      |        |
|----------------------------------------------------------------------|------|--------|
| survalis - Interpretable Survival Machine Learning Framework         | CRAN | v0.7.1 |
| <hr/>                                                                |      |        |
| * 19 learners, 7 evaluation metrics, 8 interpretability methods      |      |        |
| * Maintainer                                                         |      |        |
| funcml - Functional Machine Learning Framework                       |      | active |
| <hr/>                                                                |      |        |
| * Unified framework for functional data learning in R                |      |        |
| * Maintainer                                                         |      |        |
| survdsn - Deep Neural Networks for Survival Analysis (R torch)       |      | active |
| <hr/>                                                                |      |        |
| * Cox, AFT, and discrete-time neural survival models                 |      |        |
| * Maintainer                                                         |      |        |
| tvrmst - Time-Varying RMST from Survival Matrices                    |      | active |
| <hr/>                                                                |      |        |
| * Dynamic RMST summaries for individualized survival curves          |      |        |
| * Maintainer                                                         |      |        |
| unsurv - Unsupervised Clustering of Individualized Survival Curves   |      | active |
| <hr/>                                                                |      |        |
| * Phenotyping patients from predicted survival curves                |      |        |
| * Maintainer                                                         |      |        |
| mcstatsim - Monte Carlo Statistical Simulation Tools                 |      | active |
| <hr/>                                                                |      |        |
| * Functional approach to simulation studies in R                     |      |        |
| * Maintainer                                                         |      |        |
| missCforest - Ensemble Conditional Trees for Missing Data Imputation |      | active |
| <hr/>                                                                |      |        |
| * Missing data imputation using conditional forest ensembles         |      |        |
| * Maintainer                                                         |      |        |

## Selected Publications

|                                                                                                                                       |      |
|---------------------------------------------------------------------------------------------------------------------------------------|------|
| Interpretable clinical decision support in high-risk pregnancy: A scoping review - BMC Pregnancy and Childbirth                       | 2026 |
| <hr/>                                                                                                                                 |      |
| * El Badisy I., Assarag B., Belrhiti Z.                                                                                               |      |
| Differentiating latent from active tuberculosis through activation phenotypes and chemokine markers - Biomarker Insights              | 2025 |
| <hr/>                                                                                                                                 |      |
| * Kadi C., Ahmadi N., Houdou A., El Badisy I., et al.                                                                                 |      |
| Multi-metric comparison of ML imputation methods: application to breast cancer survival - BMC Medical Research Methodology, 24(1):191 | 2024 |
| <hr/>                                                                                                                                 |      |
| * El Badisy I., Graffeo N., Khalis M., Giorgi R.                                                                                      |      |
| Risk factors for colorectal cancer survival in Morocco: interpretable ML approach - Scientific Reports, 14(1):3556                    | 2024 |
| <hr/>                                                                                                                                 |      |
| * El Badisy I., BenBrahim Z., Khalis M., et al.                                                                                       |      |
| Interpretable ML for forecasting air pollution: a systematic review - Aerosol and Air Quality Research, 24(1):230151                  | 2024 |
| <hr/>                                                                                                                                 |      |
| * Houdou A., El Badisy I., Khomsi K., et al.                                                                                          |      |
| Private hospitals in LMICs: a typology using cluster methods Morocco - BMC Health Services Research, 24(1):1231                       | 2024 |
| <hr/>                                                                                                                                 |      |

\* Zbiri S., Belghiti Alaoui A., El Badisy I., et al.

Greenhouse gas mitigation interventions for health-care systems - Bulletin of the World Health Organization, 102(3):159 2023

\* Blom I.M., Eissa M., Mattijsen J.C., El Badisy I., et al.

Meteorological and air quality parameters and COVID-19 in Casablanca, Morocco - International Journal of Environmental Research and Public Health, 19(9):4989 2022

\* Khalis M., Toure A.B., El Badisy I., et al.

## Teaching

Lecturer and Mentor AI for Public Health - Aix-Marseille University, SESSTIM Marseille, France 2022 Present

\* Machine learning methods, NLP, applied AI for public health

\* Master's students in public health and AI for health

Lecturer Introduction to Biostatistics - University Mohammed VI of Health Sciences (UM6SS) Casablanca, Morocco 2021 Present

\* Descriptive and inferential statistics, statistical reasoning for health sciences

\* More than 100 teaching hours annually

Lecturer Health Data Analysis and Machine Learning - Institut Mines-Télécom, IMT Atlantique Brest, France 2020 Present

\* Fundamentals of ML, quantitative epidemiology, supervised and unsupervised learning

\* Second- and third-year engineering students, 21+ teaching hours annually

## Selected Talks

Comparison of missing data imputation methods in Cox regression with time-varying effects - ISCB 2023 Rome 2023

\* El Badisy I. et al.

Missing data imputation using a meta-algorithm metaCART: simulation under MCAR - EpiClin 2023 Nancy 2023

\* El Badisy I. et al.

## Skills

Methodological: Statistical methodology ù Biostatistics and epidemiology ù Survival analysis ù Missing data methods ù Meta-analysis ù Interpretable machine learning ù Computational biostatistics ù High-performance computing (Slurm)  
Technical: R (statistical and ML package development) ù Python ù Julia (learning) ù Quarto / R Markdown / LaTeX ù Git and GitHub ù Reproducible research workflows